MAHIR AFSER PAVEL

AI Researcher | Medical AI · Multimodal Learning · Efficient and Reliable Deep Learning

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Google Scholar · ORCID · ResearchGate · GitHub · LinkedIn · Portfolio

Profile Summary

AI researcher specializing in **medical image analysis**, with an emphasis on developing computationally efficient, reliable, and interpretable AI methods. Research spans knowledge distillation, explainable AI, vision-language models, quantization, federated learning, and visual question answering. **First author** of articles in Scientific Reports and PLOS One; **co-first author** of an article in the International Journal of Dentistry; second author of articles in Drones and Vehicles; and third author of an article in IEEE Access. Google Scholar profile: **100+ citations, h-index: 3,** and **i10-index: 2,** verified (November 2026).

Research Interests

- **Medical Image Analysis and Clinical AI:** Medical image classification and segmentation, disease detection, and AI-assisted clinical triage.
- **Multimodal Learning and Foundation Models:** Vision-language models (VLMs), medical image-text alignment, and visual question answering (VQA).
- **Efficient and Explainable AI:** Knowledge distillation, model compression and quantization, explainable AI (XAI), and computationally efficient deep learning.
- **Clinical Natural Language Processing and Privacy-Preserving AI:** Federated learning, generative augmentation, clinical text analysis, robustness, and interpretable AI

Education

North South University—Dhaka, Bangladesh
Bachelor of Science in Computer Science and Engineering | 2019–2023

- **CGPA: 3.71/4.00** [*Grading Policy*]
- ***CSE-specific GPA: 3.91****/4.00 (Capstone Design, Core, and Elective Courses)*
- ***Class Rank: 38th*** *out of 595 students*

B.Sc. Thesis: *"Non-small cell lung cancer detection through knowledge distillation approach with teaching assistant"*

- **Supervisor:** Dr. Riasat Khan (Associate Professor, Department of ECE). *[Link]*
- **Summary:** Implemented a three-stage knowledge distillation architecture for lung cancer classification on the *NSCLC-Radiomics* dataset, achieving 94.53% accuracy, and later deployed it in a user-friendly web application.

Scholarly Publications

Peer-Reviewed Journal Publications

- ***Multi-stage knowledge distillation with layer fusion-based deep learning approach for skin cancer classification***.
  - ✔ ***Authors: Mahir Afser Pavel****, Ramisa Asad, Goh Kah Ong Michael, Md Ikramuzzaman, Murad Mustakim, and Riasat Khan.*
  - ✔ ***Venue****: Scientific Reports (Nature Portfolio), 2025. DOI: 10.1038/s41598-025-23403-2 [Code]*
- ***Non-small cell lung cancer detection through knowledge distillation approach with teaching assistant***
  - ✔ ***Authors: Mahir Afser Pavel****, Rafiul Islam, Shoyeb Bin Babor, Riaz Mehadi, and Riasat Khan*
  - ✔ ***Venue****: PLOS One (PLOS), 2024. DOI: 10.1371/journal.pone.0306441 [Code]*
- ***Error-Tolerant Multimodal Vision-Language Models for Endodontic Triaging: A Cross-Sectional Study***.
  - ✔ ***Authors: Mahir Afser Pavel*** *(Co-First Author), Md Fahim Shahoriar Titu (Co-First Author), Afifa Zain Apurba, Saif Ahmed, Shafin Rahman, James Dudley, and Taseef Hasan Farook*
  - ✔ ***Venue****: International Journal of Dentistry (Wiley), 2026. DOI: 10.1155/ijod/4148741. [Code]*
- ***Fusion of Image Filtering and Knowledge-Distilled YOLO Models for Root Canal Failure Diagnosis***
  - ✔ ***Authors:*** *Afifa Zain Apurba, Md Fahim Shahoriar Titu,* ***Mahir Afser Pavel****, Intisar Tahmid Naheen, and Riasat Khan*
  - ✔ ***Venue:*** *IEEE Access (IEEE), 2025. DOI: 10.1109/ACCESS.2025.3560998. [Code]*
- ***Real-time fire detection: Integrating lightweight deep learning models on drones with edge computing***
  - ✔ ***Authors:*** *Md Fahim Shahoriar Titu,* ***Mahir Afser Pavel****, Goh Kah Ong Michael, Hisham Babar, Umama Aman, and Riasat Khan.*
  - ✔ ***Venue****: Drones (MDPI), 2024. DOI: 10.3390/drones8090483 [Code]*
- ***Hybrid ViT-RetinaNet with Explainable Ensemble Learning for Fine-Grained Vehicle Damage Classification***
  - ✔ ***Authors****: Ananya Saha,* ***Mahir Afser Pavel****, Md Fahim Shahoriar Titu, Afifa Zain Apurba, and Riasat Khan.*
  - ✔ ***Venue:*** *Vehicles (MDPI), 2025. DOI: 10.3390/vehicles7030089. [Code]*

Manuscripts Under Review

- ***Federated Multi-Component Framework for Brain Tumor Diagnosis with Generative Augmentation and Explainable AI***
  - ***Authors: Mahir Afser Pavel****, Nafiz Fahad, Md Tanzib Hosain, Md. Kishor Morol, and Tze Hui Liew*
  - ***Status:*** *Under Review at Array (Elsevier), 2026*
- ***CLKD-MED: A novel cross-lingual knowledge distillation framework for multilingual clinical outcome prediction***
  - ***Authors: Mahir Afser Pavel****, Rafiul Islam, Mohammad Junayed Hasan, and M. R. C. Mahdy*
  - ***Status:*** *Under Review at Information Processing & Management (Elsevier), 2026*

- ***SLGRA-Net: Structure-Aware Latent Graph Reasoning for 3D Brain Tumor Segmentation***
  - ***Authors: Mahir Afser Pavel****, Nafiz Fahad, Md Tanzib Hosain, Md. Kishor Morol, and Tze Hui Liew*
  - ***Status:*** *Under Review at Biomedical Signal Processing and Control (Elsevier), 2026*

Research in Progress

- ***Trustworthy Clinical AI for Diabetic Retinopathy Grading via Hybrid Foundation Models, Calibration-Aware Uncertainty Learning, and Multi-Criteria Clinical Referral***
- ***S³-Net: Self-Stabilizing, Self-Regulating, and Self-Optimizing Neural Networks via Universal Stability Learning***
- ***FULCRUM: Option-Contrastive Frame Selection for Long-Video Understanding***
- ***ACP-KD: Adaptive Clinical Prototype Knowledge Distillation with Dynamic Feature Fusion and Confidence-Aware Multi-Objective Learning for Lightweight Skin Lesion Classification***

Professional Experience

Research Assistant | ELITE Research Lab LLC (Dhaka, Bangladesh) (Aug 2025 – Present)

- **Federated Brain Tumor Classification:** Developed *Fed-GAN-DE-CL*, a federated multi-component framework that integrates generative adversarial networks and dual-encoder contrastive learning for decentralized brain tumor classification without centralizing patient data.
- **3D Brain Tumor Segmentation:** Proposed *SLGRA-Net*, incorporating structure-aware latent graph reasoning to model spatial relationships and address ambiguous tumor boundaries in 3D MRI segmentation.

Research Assistant | United International University (Dhaka, Bangladesh) (May 2025 – Jun 2025)

- **Experimental Pipeline Development:** Designed and executed data preprocessing, feature engineering, and model-training workflows for deep learning and natural language processing research.
- **Literature Review and Research Proposals:** Synthesized research on emerging multi-agent systems and contributed to drafting research proposals.

Research Assistant | North South University (Dhaka, Bangladesh) (May 2024 – Jul 2024)

- **Edge AI and Real-Time Fire Detection:** Co-developed a drone-based fire detection pipeline using knowledge distillation to deploy lightweight YOLOv8n models on Raspberry Pi 5 devices for edge inference.

Research Trainee | Mahdy Research Academy (Remote / Online) (Jan 2024 – Mar 2025)

- **Cross-Lingual Clinical NLP:** Introduced *CLKD-MED*, a knowledge distillation framework for multilingual clinical outcome prediction using MIMIC-III 92,293 clinical notes, transferring clinical representations across English and lower-resource languages to predict length of stay and mortality.

Teaching & Research Experience

- **Research Leadership:** Led research design and development of deep learning, computer vision, and clinical NLP pipelines at *ELITE Research Lab LLC* and *Mahdy Research Academy.*
- **Academic Peer Review:** Served as a peer reviewer for *IEEE Access* and *Heliyon*, evaluating research manuscripts for methodological rigor, technical quality, and clarity.
- **Teaching and Student Mentoring:** Co-led academic sessions and supported laboratory instruction in *Digital Electronics, Analog Electronics, and Senior Design at* North South University, mentoring student groups on circuit mapping, hardware–software co-design, and research manuscript preparation under faculty supervision.

Standardized Examination Scores

IELTS Academic (2026): Overall: 6.5 | Listening: 7.0 | Reading: 7.0 | Writing: 6.5 | Speaking: 6.0

Academic Projects & Research Focus

- **Quantization-Aware Edge AI:** Evaluating lightweight deep learning models (YOLOv5 and YOLOv8) and multimodal vision-language models (BLIP, CLIP, Florence-2, and PaliGemma), exploring Low-Rank Adaptation (LoRA), and quantization-aware training to reduce model size and memory requirements for resource-constrained clinical applications.
- **Medical Image Analysis & Knowledge Distillation:** Developing multi-stage teacher–assistant–student distillation pipelines using Vision Transformers (ViT) and ResNet152v2 for medical image analysis, including skin-lesion classification on HAM10000 and lung cancer analysis using NSCLC CT data, with a focus on diagnostic performance and model efficiency.
- **Privacy-Preserving Biomedical Machine Learning:** Investigating federated generative adversarial networks and contrastive learning methods to support collaborative model training across decentralized clinical datasets while reducing the need to centralize sensitive patient data.

Technical & Research Skills

Category	Tools and Technologies
Programming Languages	Python (Advanced), C++, Java, Bash
AI/ML Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV
Generative AI & VLMs	LangChain, LLaMA, RAG, LoRA, QAT, BLIP, CLIP, Florence-2, PaliGemma
Research & Development Tools	Git, GitHub, LaTeX, Overleaf
Databases & Analysis	SQL, MySQL, NumPy, Pandas, Matplotlib, SPSS, Microsoft Excel

Leadership & Extracurricular Activities

- **Research Team Coordinator (2023–2024):** Coordinated a team of five research assistants across senior design and directed research projects, supporting data-pipeline development and manuscript preparation under faculty mentorship. Contributed to research activities associated with five peer-reviewed journal articles.
- **Manuscript Reviewer (2026):** Reviewed research manuscripts for methodological rigor, technical quality, and clarity as part of the academic peer-review process.
- **Academic Peer Mentor (2022–2023):** Mentored junior engineering students in laboratory settings, helping troubleshoot hardware–software interfacing issues and supporting scientific writing workshops.

Honors & Awards

- **Magna Cum Laude (December 2024):** Awarded by North South University for academic excellence in the B.Sc. in Computer Science and Engineering program; ranked within the top 7% of the cohort.
- **Academic Merit Scholarships:** Received multiple tuition waivers of 10%–20% from North South University based on semester academic performance, with semester GPAs reaching 4.00.
- **Peer Reviewer Recognition (March 2026):** Recognized for peer-review contributions to IEEE Access, Heliyon, and Frontiers in Computer Science.

References

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